DE–7382 **11**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, DECEMBER 2013.

Chemistry

ORGANIC CHEMISTRY — I

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) What is meant by hyperconjugation? Explain about the steric inhibition of resonance. (6)

   (b) Explain the following :

   (i) Homoaromaticity and antiaromaticity. (6)

   (ii) Alternate hydrocarbons. (4)

   (iii) Aromaticity of fulvene and azulene. (4)

2. (a) Write briefly about the mechanism of Gatermann Koch formylation and Von-Richter reaction. (8)

   (b) Describe the effect of solvent polarity and the structure of the substrate on the course of the aliphatic nucleophilic substitution reactions. (7)

   (c) Discuss about the benzyne mechanism. (5)

3. (a) Outline the significance of E_2 and E_1CB mechanisms. (5)

   (b) Write notes on Diels-Alder reaction, Darzen glycidic ester synthesis and Michael addition. (9)

   (c) Explain the factors which are influencing the addition to conjugated systems. (6)

4. (a) Write briefly about the methods of preparing flavones and indole. (6)

   (b) Explain the biosynthetic routes of terpenes. (6)

   (c) Describe the structure and synthesis of morphine. (8)

5. (a) Establish the structure of oxytocin. (8)

   (b) Write notes on :

      (i) End group analysis. (4)

      (ii) Primary structure of proteins. (4)

      (iii) Enzymes as catalysts. (4)

6. (a) Discuss the formation and stability of reactive intermediates. (8)

   (b) Explain the aromaticity of the various types of annulenes. (6)

   (c) Describe the kinetic and thermodynamic control of chemical reactions. (6)

7. (a) Enumerate the significance of orientation and reactivity in aromatic nucleophilic substitution reaction. (6)

   (b) Explain Bredt's rule and pyrolytic cis elimination.(6)

   (c) Write the mechanism of Dickmann condensation and Wittig reaction. (8)

8. (a) Establish the synthetic routes of cis-norpinic acid and camphoronic acid. (4)

   (b) Explain the configuration and conformation of disaccharides. (6)

   (c) Indicate how α-pinene and atropine can be synthesized. (10)

DE–7383 **12**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION,
DECEMBER 2013.

INORGANIC CHEMISTRY — I

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Describe the magnetic behaviour of B_2 and C_2 molecules. (6)

   (b) What is Madelung constant? How is it evaluated? (4)

   (c) Discuss the theoretical basis of hard and soft acid and bases. (5)

   (d) Discuss the formation of bonding molecular orbital in homonuclear diatomic molecules. (5)

2. (a) Outline the difference in the crystal structure of NaCl and CsCl and of CaF_2 and TiO_2. (6)

   (b) Explain how doping and imperfections can lead to semi conductivity in solids. (7)

   (c) Explain Schottky and Frenkel defects in stoichiometric crystals. (7)

3. (a) Discuss the principles and applications of isotope dilution analysis. (8)

(b) Write notes on

(i) nuclear cross section (6)

(ii) cyclotron. (6)

4. (a) Discuss the principle of solvent extraction show how it can be applied in the separation of plutonium form uranium fuel element of a nuclear reactor. (7)

(b) Comment on the magnetic properties of the tripositive lanthanides. (6)

(c) Compare and contrast the lanthanides and actinides in their oxidation states other than +3. Account for their difference. (7)

5. (a) Distinguish isopoly anions from heteropoly anions. Discuss briefly about Keggin's work on 12 polyacids. (7)

(b) Discuss the different types of structures found in silicates. (7)

(c) The hardness of layer silicates increases in the order pyrophyllilte<muscovite<margarite. Explain. (6)

6. (a) Compare and contrast VB and MO theories of bonding in molecules. (6)

(b) Define lattice energy. Taking NaCl crystal as example describe Born–Meyer approach to the calculation of lattice energy of ionic crystals. (6)

(c) Discuss the structural features of

(i) Wurtzite and

(ii) Cadmium iodide. (8)

7. (a) Explain secular equilibrium among radioactive elements. (5)

(b) Write an account of the shell model of the nucleus. (5)

(c) Comment on nuclear reactions responsible for solar energy. (6)

(d) Distinguish between intrinsic and extrinsic semiconductivity. (4)

8. (a) Compare lanthanides and actinides with reference to their magnetic and spectral properties. (8)

(b) Discuss the structure of polymolybdates and vanadates. (6)

(c) Bring out the differences and similarities between

(i) Pynoxenes and amphiboles

(ii) Zeolites and ultramarines. (6)

DE–7384 **13**

DISTANCE EDUCATION

M.Sc.(Chemistry) DEGREE EXAMINATION,
DECEMBER 2013.

PHYSICAL CHEMISTRY – I

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Explain the determination of partial molar volume for a binary system. (5)

   (b) How can you test the validity of third law of thermodynamics? (5)

   (c) How will you determine the fugacity of real gas by graphical method? State how can it vary with pressure. (10)

2. (a) Derive the expression for the entropy change accompanying variation of

      (i) Temperature and volume

      (ii) Temperature and pressure. (12)

   (b) Derive any one thermodynamic equation of state. (4)

   (c) Write note on thermodynamic properties at absolute zero. (4)

3. (a) Write a note on transport number and ionic mobility. (4)

(b) Show that $\log \gamma_{\pm} = -0.509 \left| Z^{+} Z^{-1} \right| I^{\frac{1}{2}}$ (10)

(c) What is over voltage and explain how its be prevented? (6)

4. (a) State and explain the term quantum yield. How do you account for the fact that the quantum yield of the photochemical reaction $H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$ is low while that of the reaction $H_{2_{(g)}} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$ is very high? (7)

(b) Write note on :

(i) Operators

(ii) Matrices

(iii) Vectors. (3 + 3 + 3)

(c) State the importance of Heisenberg's uncertainty principle. (4)

5. (a) Set up and solve Scrodinger equation for a particle in a 3-D box to get eigen functions and eigen values. (8)

(b) Discuss the postulates of quantum mechanics. (6)

(c) Derive Schrodinger wave equation. (6)

6. (a) For the reaction $A + B \underset{u_2}{\overset{u_1}{\rightleftharpoons}} C$ the following data were obtained :

$[A]_0 = 0.1164$ mole/lit $[B]_0 = 0.0758$ mole/lit

$[A]_e = 0.0772$ mole/lit.

If [C] after 73.33 minutes is 0.0711 mole/lit, calculate K_1 and K_2. (6)

(b) Derive and discuss the transition state theory and illustrate its applications. (10)

(c) State and explain KRRM theory. (4)

7. (a) What is chemical actinometry? What are the uses of it? Explain the different types of chemical actinometers. (8)

(b) What is solar energy? How will you harnessing the solar energy? Mention its demerits. (7)

(c) Explain the photophysical process in electronically excited state. (5)

8. (a) Discuss the study of fast reactions using relaxation and NMR methods. (12)

(b) What is steady state approximation? How is it applicable in chemical reaction? (4)

(c) Explain the mechanism involved in photo decomposition of Carbonyl compounds. (4)

———————

DE–7385 **14**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, DECEMBER 2013.

Chemistry

INSTRUMENTAL METHODS OF ANALYSIS

(2008 onwards)

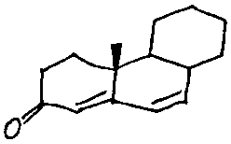
Time : Three hours Maximum : 100 marks

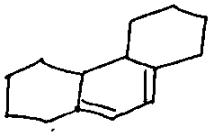
Answer any FIVE questions.

Each questions carries 20 marks. ($5 \times 20 = 100$)

1. (a) Discuss the factors which are affecting the position and intensity of absorption bands in uv-visible spectroscopy. (10)

(b) Predict the λ_{max} value for the following compounds (5)

(i) 

(ii) 

(c) An organic compound (A) with molecular formula C_3H_9N shows the following peaks in the IR spectrum. 3012 $cm^{-1}_{(m)}$, 3423 $cm^{-1}_{(s)}$, 3236 $cm^{-1}_{(m)}$, 1615 $cm^{-1}_{(m)}$ when the compound (A) is treated with HNO_2 acid we get a compound (B) which shows a strong peak at 3430 cm^{-1}, what are (A) and (B) and explain the reactions involved? (5)

2. (a) What is chemical shift? Explain various factors which are affecting the chemical shift. (7)

   (b) Write notes on :

      (i) Spin tickling

      (ii) Nuclear overhauser effect. (10)

   (c) Explain why TMS is used as a standard in NMR spectroscopy. (3)

3. (a) Explain the terms :

      (i) α - halo ketone rule.

      (ii) Mclafferty rearrangement. (10)

   (b) Explain how paper chromatographic analysis is carried out. (6)

   (c) Discuss the fragmentation pattern of hexane and write various peak values obtained in Mass spectroscopy. (4)

4. (a) Discuss the principle and applications of calorimetric technique. (10)

   (b) Discuss the working principle of thermo Gravimetric analysis. (10)

5. (a) Write the principle and applications of polarography. (10)

   (b) Write a note on linear regression analysis. (10)

6. (a) Write a note on over tones and combination bands with examples. (8)

   (b) Explain Fermi resonance. (8)

   (c) Cyclopentanone (0.43 M in a 1.0 cm cell) has λ_{max} at 288 nm with an absorbance of 0.85. What is the Molar absorptivity of cyclopentanone at 288 nm? (4)

7. (a) Explain the advantages of FT-pulse technique in recording NMR spectra. Explain the principle in detail. (10)

   (b) Discuss the principles and working technology of High Pressure Liquid Chromatography (HPLC). (10)

8. (a) Write notes on :

   (i) Turbidimetry.

   (ii) Flame photometry. (10)

   (b) Explain the applications of cyclic voltammetry. (10)

DE–7386 **21**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION,
DECEMBER 2013.

ORGANIC CHEMISTRY — II

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) How would you decide whether a rearrangement process is inter-molecular or intramolecular? (6)

   (b) What is DCC? Give any two reactions which show the synthetic utility of DCC. (4)

   (c) Discuss the use of periodic acid in oxidation. Give a suitable mechanism. (4)

   (d) Discuss the following :

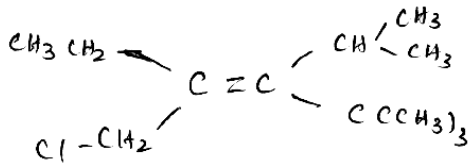
      (i) Robinson annelation sequence.

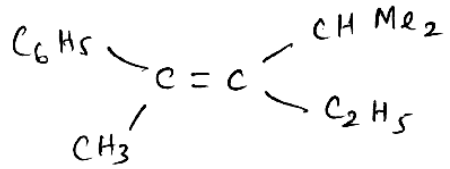
      (ii) Retro-synthetic analysis. (3 + 3)

2. (a) Write a brief note on asymmetric synthesis. (4)

   (b) State the conditions required for a compound to exhibit optical activity. (4)

(c) Assign E or Z notation for the following : (4)

(i) 

(ii) 

(d) Describe the method for the determination of configuration of ketoximes. (8)

3. (a) Explain neighboring group participation with suitable examples. (6)

(b) Discuss the stability of triaryl radicals. (6)

(c) Discuss the mechanism of Haudsdicker reaction. Give examples. (4)

(d) Differentiate configuration and conformation. (4)

4. (a) Allylic and benzylic radicals are more stable and less reactive – rationalize. (4)

(b) Discuss the optical activity of biphenyls. (4)

(c) Write down the preferred conformation of the active form of butane – 2, 3, – diol. (4)

(d) Using Cram's rule predict the product formed by the reaction of α-phenyl propionaldehyde with methyl magnesium bromide. (4)

(e) Give brief account of the following stability difference between trans and cis decaline. (4)

5. (a) Write notes on :

   (i) MPV reduction.

   (ii) Selenium dioxide oxidation. (4 + 4)

   (b) Distinguish between relay and convergent approaches. (4)

   (c) Write any two methods of resolution of racemic mixtures. (8)

6. (a) Explain Norrish type-I photochemical reaction with two examples. (8)

   (b) Write briefly on :

   (i) Sigmatropic rearrangement.

   (ii) Photo reduction.

   (iii) di-pi-methane rearrangement. (4 + 4 + 4)

7. (a) Describe briefly about quantum efficiency. (5)

   (b) Give the mechanism of paterno-Buchi reaction. (5)

   (c) Elucidate the structure of Vitamin – A. (10)

8. (a) Outline the synthesis of equilenin. (5)

   (b) Explain how the position of the angular methyl groups established in cholesterol. (5)

   (c) How is the structure of Vitamin – D_2 established? (10)

DE–7387 **22**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION,
DECEMBER 2013.

INORGANIC CHEMISTRY — II

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Why is the complex $(Ni\,cl_4)^{2-}$ paramagnetic while $Ni(Cn)_4)^{2-}$ is diamagnetic? (6)

   (b) Discuss the various factors influencing the magnitude of crystal field splitting. (6)

   (c) Draw and explain the molecular orbital energy level diagram of $(Co\,f_6)^{3-}$. (8)

2. (a) Explain the spectrophotometric method of determination of stability constant of a metal complex. (6)

   (b) Write a note on Template effect and Macrocyclic ligands. (6)

   (c) Write briefly on the tunneling mechanism of electron transfer reactions, citing suitable illustration. (8)

3. (a) Explain how epr spectral measurements are useful in studying John Teller effects in Cu(11) complexes. (6)

(b) What are orgel diagrams? How are they differ from Sugano – Tanabe diagrams? (6)

(c) Explain with examples the different types of magnetic behaviour in inorganic systems. (8)

4. (a) Discuss the structure and bonding in metal – olefin complexes. (6)

(b) Discuss the mechanism of hydrogenation of alkenes using Wilkinson's catalyst. (8)

(c) How will you establish the aromatic character of cyclopentadienyl groups in ferrocene? (6)

5. (a) Discuss the mode of action of carboxy peptidase in the hydrolysis of an amide linkage in a polypeptide. (6)

(b) Explain Chelate therapy. (6)

(c) Discuss the structure and functions of Vitamin B_{12}. (8)

6. (a) Describe the splitting of d–orbitals for a square planar complex and account for the diamagnetic nature of nickel dimethyl – glyoximate complex. (8)

(b) Calculate the crystal field stabilisation energy in terms of Dq for $(Fe(CN)_6)^{4-}$ and $(Fe\ Cl_6)^{3-}$ ions. (6)

(c) List out the factors which affect the stability of a metal complex. (6)

7. (a) How is trans effect employed for distinguishing between cis and trans isomers of $[Pt\ A_2X_2]^o$ type complexes? (6)

(b) Construct the Orgel diagram for a d^9 configuration and explain the hole formalism. (6)

(c) What is Zeigler – Natta catalyst? Describe propylene polymerisation. (8)

8. (a) Explain how the symmetry of the complex affects the number of absorption bands in IR? (6)

(b) Discuss the structure and important function of Cytochrome C. (6)

(c) Explain briefly:

(i) Invivo nitrogen fixation and

(ii) Sodium ion pump. (8)

DE–7388 **23**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION,
DECEMBER 2013.

PHYSICAL CHEMISTRY –II

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Derive an expression for an Internal energy partition function. (8)

   (b) Give the statistical interpretation of third law of thermodynamics. (6)

   (c) What are the limitations of Maxwell-Boltzmann statistics? (6)

2. (a) Set up Schrodinger wave equation for hydrogen atom and obtain its solution. (16)

   (b) What will happen if the walls of one dimensional box are suddenly removed? (4)

3. (a) Construct the character table for C_3V point group using postulates of great orthogonality theorem. (10)

   (b) Explain the application of group theory to electronic spectra with suitable example. (10)

4. (a) What are the factors that influence the reaction rate in solution? (8)

   (b) Discuss the application of Hammett and Taft equation in the study of substituent effects on the rate of chemical reactions. (12)

5. (a) What are the factors influencing adsorption? (8)

   (b) Derive Langmuir adsorption isotherm and explain its success and limitations. (12)

6. (a) What are the limitation of Einstein Theory and Debye Theory of specific heat? (6)

   (b) Write note on Pauli's exclusion principle. (6)

   (c) Listout the rules for forming group and group multiplication table. (8)

7. (a) Derive an expression for Boltzmann statistics. (6)

   (b) Discuss the application of perturbation method to Helium. (6)

   (c) Write note on primary and secondary salt effects. (8)

8. (a) Define the following terms :

      (i) Symmetry elements

      (ii) Symmetry operation. (4)

   (b) Describe acid-base catalysis with suitable example. (4)

   (c) What are the postulates of B.E.T. equation? Derive B.E.T. equation. (12)

DE–7389 **24**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, DECEMBER 2013.

APPLIED CHEMISTRY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Suggest any four methods to prevent radioactive pollution. (8)

   (b) Describe the harmful effects of water pollution and the methods used to control it. (6)

   (c) Explain the preliminary treatment of waste water. (6)

2. (a) What is ozone umbrella? How is ozone formed in stratosphere? How is ozone layer depleted? Explain the mechanism of ozone depletion. What are the effects of ozone depletion? (10)

   (b) List out the hazardous materials and indicate their ill effects. (5)

   (c) Give an account of biological investigation of water. (5)

3. (a) How will you measure the corrosion rate using electrochemical and chemical methods? Describe the techniques briefly. (8)

   (b) Discuss the principles and applications of alloy plating of brass. (7)

   (c) Define Electroplating. Mention its advantages. (5)

4. (a) Describe the principles, advantages and limitations of electroplating. (10)

   (b) Explain the principles and applications of Brush plating. (6)

   (c) Write note on cathodic protection of corrosion. (4)

5. (a) How are the polymers obtained from suspension and emulsion polymerization techniques? (9)

   (b) Illustrate briefly about the mechanism of cationic polymerization. (6)

   (c) Explain the functionality of monomers and its significance. (5)

6. (a) What are electronically conducting polymers? How are they produced? Discuss the electrical conductivity of poly sulphur nitride and poly metylene. (7)

   (b) Discuss the techniques of bulk polymerization. (7)

   (c) What is anodizing? Explain the principles and coloring of anodizing aluminium. (6)

7. (a) Describe the synthesis of nanoparticles using solvothermal method. How it can be characterized using XRD and XPS techniques? (12)

   (b) Give an account of spray pyrolysis and electrochemical deposition of nano particles. (8)

8. (a) What is internet? Who provide the internet service? How can you access the internet? Using internet how will you send an e-mail and use the search engines for various purpose? (12)

   (b) Enumerate the source of journals and chemical abstracts. Explain how will you get the articles with help of internet services. (8)